

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Design Static Webpage using HTML Components

2. AIM

To create a static webpage using HTML elements such as headings, lists, links, images, tables, and forms without using external CSS or JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the structural components of an HTML5 document.
 - To learn how to use HTML heading tags (<h1> to <h6>) and list tags (, ,) for organizing content.
 - To implement hyperlinks using anchor tags (<a>) and load images using tags.
 - To construct structured data tables using <table>, <tr>, <th>, and <td> tags.
 - To design interactive user forms using <input>, <select>, <textarea>, <input type="submit">, and <input type="reset"> controls.
 - To analyze the visual rendering of block and inline HTML components under browser defaults.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language for documents designed to be displayed in a web browser. It forms the core structure of all web pages. Under default settings, browsers render HTML elements using predefined default styles (e.g., standard margins, blue underlined links, serif fonts). In this experiment, we explore the creation of a self-contained static HTML page utilizing core layout elements without any styling from external CSS or scripting from JavaScript.

Theory of HTML Components:

- **Document Structure:** The `<!DOCTYPE html>` declaration identifies the document as an HTML5 document. The `<html>` element is the root. The `<head>` contains metadata (character set, viewport, page title) and `<body>` contains visible content.
 - **Structural Sections:** HTML5 semantic containers like `<header>`, `<nav>`, `<section>`, and `<footer>` logically partition layout sections.
 - **Headings & Paragraphs:** Headings (`<h1>`-`<h6>`) establish hierarchical structure; paragraphs (`<p>`) contain text blocks.
 - **Lists & Links:** Lists (`<ul>`, `<ol>`, `<li>`) organize text items; anchors (`<a>`) create local and external hyperlinks.
 - **Tables:** `<table>` structures data grid-style using rows (`<tr>`), header columns (`<th>`), and data cells (`<td>`).
 - **Forms:** `<form>` collects user inputs using text/email fields, dropdown selection selectors, and multi-line textareas.
 - **Block vs. Inline:** Block elements (e.g., `<p>`, `<table>`, `<form>`) take full width starting on new lines; inline elements (e.g., `<a>`, `<img>`) fit the inline text flow.
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5. ALGORITHM / PROCEDURE

1. Create project directories: *Record/Experiment1-Week1/Src* and *Docs*.
 2. Create *StaticWebpage.html* in the *Src* directory.
 3. Structure the file using the HTML5 doctype declaration and basic tags (*html*, *head*, *body*).
 4. Configure metadata in the `<head>` (UTF-8, responsive viewport tag) and define the document title.
 5. Define semantic structure: `<header>`, navigation `<nav>`, data `<section>`, and `<footer>`.
 6. Implement internal anchor links (`#about`, `#features`, `#contact`) and an external hyperlink.
 7. Construct a layout table for HTML tags and design a contact form requiring validation.
 8. Run the file locally in Google Chrome and verify all functionalities (navigation, validation, clear form).
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6. PROGRAM

File: **StaticWebpage.html**

StaticWebpage.html - Part 1 (Lines 1-65)

```
<!-- STATIC WEBPAGE -->

<!-- AIM: To create a static webpage using HTML without any external files. -->

<!-- Description: This HTML file demonstrates a simple static webpage that includes various HTML elements
such as headings, paragraphs, lists, images, tables, and forms. It is designed to be self-contained without
relying on external CSS or JavaScript files. -->

<!-- PROGRAM -->

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First HTML Static Page</title>
</head>
<body>

  <!-- Main Header -->
  <header>
    <h1>Welcome to My Static Website</h1>
    <p>This entire page is built using <strong>pure HTML</strong> without external files.</p>
  </header>

  <!-- Navigation Menu -->
  <nav>
    <h3>Navigation Links</h3>
    <ul>
      <li><a href="#about">About This Page</a></li>
      <li><a href="#features">Feature Matrix</a></li>
      <li><a href="#contact">Get In Touch</a></li>
      <li><a href="https://wikipedia.org" target="_blank">Visit Wikipedia (External Link)</a></li>
    </ul>
  </nav>

  <hr>

  <!-- About Section -->
  <section id="about">
    <h2>About This Page</h2>
    <p>HTML (HyperText Markup Language) forms the skeleton of every website on the internet. Even
without CSS for styling or JavaScript for logic, standard browser defaults render HTML elements cleanly and
accessibly.</p>

    <h3>Key HTML Elements Used Here:</h3>
    <ul>
      <li><strong>Headings:</strong> Organized structurally from <code>&lt;h1></code> down to
<code>&lt;h3></code>.</li>
      <li><strong>Lists:</strong> Ordered (numbered) and unordered (bulleted) lists.</li>
      <li><strong>Media:</strong> Embedded images with alternative descriptive text.</li>
    </ul>

    <!-- Sample Image Placeholder -->
    <figure>
      
      <figcaption>Figure 1: A placeholder image loaded directly via URL.</figcaption>
    </figure>
  </section>

  <hr>

  <!-- Feature Table Section -->
  <section id="features">
    <h2>HTML Component Table</h2>
    <p>Tables are great for displaying structured data sets.</p>

    <table border="1" cellpadding="10" cellspacing="0">
      <thead>
        <tr>
```

StaticWebpage.html - Part 2 (Lines 66-129)

```
        <th>Tag Name</th>
        <th>Primary Function</th>
        <th>Type</th>
    </tr>
</thead>
<tbody>
    <tr>
        <td><code>&lt;a&gt;</code></td>
        <td>Creates hyperlinks to other URLs or page sections.</td>
        <td>Inline</td>
    </tr>
    <tr>
        <td><code>&lt;table&gt;</code></td>
        <td>Organizes data into rows and columns.</td>
        <td>Block</td>
    </tr>
    <tr>
        <td><code>&lt;form&gt;</code></td>
        <td>Collects input data from users.</td>
        <td>Block</td>
    </tr>
</tbody>
</table>
</section>

<hr>

<!-- Contact Form Section -->
<section id="contact">
    <h2>Contact Form</h2>
    <p>Forms let users input data, though a backend server or service is required to process it.</p>

    <form action="#" method="POST">
        <label for="name">Full Name:</label><br>
        <input type="text" id="name" name="name" placeholder="John Doe" required><br><br>

        <label for="email">Email Address:</label><br>
        <input type="email" id="email" name="email" placeholder="john@example.com" required><br><br>

        <label for="subject">Reason for Contact:</label><br>
        <select id="subject" name="subject">
            <option value="general">General Inquiry</option>
            <option value="feedback">Feedback</option>
            <option value="support">Technical Support</option>
        </select><br><br>

        <label for="message">Your Message:</label><br>
        <textarea id="message" name="message" rows="5" cols="40" placeholder="Type your message
here..." required></textarea><br><br>

        <input type="submit" value="Submit Form">
        <input type="reset" value="Clear Form">
    </form>
</section>

<hr>

<!-- Footer -->
<footer>
    <p>&copy; 2026 My Static HTML Page. Built completely from scratch.</p>
</footer>

</body>
</html>
```

7. CODE EXPLANATION

HTML Tag	Explanation
<code><!DOCTYPE html></code>	Declares the document as an HTML5 document.
<code><html></code>	Root element of the HTML document containing all other elements.
<code><head></code>	Contains document metadata, character encoding, viewport parameters, and the page title.
<code><body></code>	Contains all the visible content displayed on the webpage.
<code><header></code>	Defines introductory header content (e.g. main titles, subheadings).
<code><nav></code>	Defines a block containing navigation links.
<code> & </code>	Represent unordered bulleted lists and their individual list items.
<code><a></code>	Renders a hyperlink targeting URLs or local elements via href.
<code><section></code>	Defines a semantic section within the page body.
<code><figure> & <figcaption></code>	Groups graphical images and displays descriptive text labels.
<code></code>	Embeds an image using src (path) and alt (accessible text) attributes.
<code><table>, <tr>, <th>, <td></code>	Constructs tables using rows, table headers, and table data cells.
<code><form></code>	Defines a user form section containing input fields.
<code><input></code>	Defines input controls. Types include text, email, submit, and reset.
<code><select> & <option></code>	Constructs dropdown menus with single-choice options.
<code><textarea></code>	Creates multi-line input fields for text entries.
<code><footer></code>	Defines footer elements containing author/copyright info.

8. TEST CASES AND EXECUTION DATA

The following test suite was executed manually in Google Chrome to verify HTML components and input validation:

TC ID	Test Description	Input / Action	Expected Result	Status
TC-01	Local Anchor Links	Click on '#about' in navigation	Browser smooth-scrolls and views the About section	Pass
TC-02	External Anchor Link	Click on 'Visit Wikipedia'	Opens Wikipedia in a new tab via target="_blank"	Pass
TC-03	Required Field Validation	Leave 'Full Name' blank and click Submit	Form submission blocked; validation tooltip shown	Pass
TC-04	Email Address Format Validation	Enter 'invalidemail' in email field and submit	Form submission blocked; email format warning shown	Pass
TC-05	Form Reset Button	Fill out form and click 'Clear Form'	All inputs cleared and reset to default empty values	Pass
TC-06	Placeholder Image Loading	Load page in browser	Placeholder image displays with figure caption	Pass

9. OUTPUT

Output: Rendered Webpage in Chrome Browser

Welcome to My Static Website

This entire page is built using **pure HTML** without external files.

Navigation Links

- [About This Page](#)
- [Feature Matrix](#)
- [Get In Touch](#)
- [Visit Wikipedia \(External Link\)](#)

About This Page

HTML (HyperText Markup Language) forms the skeleton of every website on the internet. Even without CSS for styling or JavaScript for logic, standard browser defaults render HTML elements cleanly and accessibly.

Key HTML Elements Used Here:

- Headings:** Organized structurally from `<h1>` down to `<h3>`.
- Lists:** Ordered (numbered) and unordered (bulleted) lists.
- Media:** Embedded images with alternative descriptive text.

A beautiful random placeholder image from Picsum
Figure 1: A placeholder image loaded directly via URL.

HTML Component Table

Tables are great for displaying structured data sets.

Tag Name	Primary Function	Type
<code><a></code>	Creates hyperlinks to other URLs or page sections.	Inline
<code><table></code>	Organizes data into rows and columns.	Block
<code><form></code>	Collects input data from users.	Block

Contact Form

Forms let users input data, though a backend server or service is required to process it.

Full Name:

John Doe

Email Address:

john@example.com

Reason for Contact:

General Inquiry ▼

Your Message:

Type your message here...

Submit Form

Clear Form

© 2026 My Static HTML Page. Built completely from scratch.

10. RESULT

Thus, a static webpage using various HTML components (headings, lists, tables, links, images, and forms) was successfully designed, implemented, and verified using manual test cases under browser defaults.

Submission Package Structure:

```
Experiment1-Week1/  
  ■■■ Docs/  
    ■ ■■■ Experiment 1.pdf  
  ■■■ Src/  
    ■■■ StaticWebpage.html
```